# Interest Rate Volatility Surface Simulation for Risk Management

*(Working Paper)*

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## 1 Problem Setup

The implied volatility of an option on an interest rate contract is the model volatility that makes the model price match the market price of the option. Swaps, forwards and future contracts are examples of the underlying instruments. Swaptions, caps and floors are examples of the options. The quoted implied volatility is usually either the normal (Bachelier) volatility, the Black (lognormal) volatility or the CEV (Constant Elasticity of Variance) volatility. The choice depends on the model used to price the option. Converting from one form of implied volatility to another can be typically done using either closed from or semi-analytic approximations.

The price function of a simple option $C$ can be assumed a function of the form $C(v\,; F_{x,t}, k, x)$ : $v \mapsto c$, where $c$ is the option price, and $F_{x,t}, v, k, x$ are the risk neutral forward price at time $x$ of an underlying contract of tenor $t$, its implied volatility, the strike price of the option and the option's time to expiry, respectively. For at-the-money options, $k = F_{x,t}$. Let $\mathbb{C}_v$ be the space of functions $C_v : (F_{x,t}, k, x) \mapsto c$, *i.e.* $C_v(.) = C(v; .)$, which are monotonic in $v$. The implied volatility surface, of such options, can be formally defined as the map

$$\begin{aligned} V: X \times T \times K &\to \mathbb{C}_v \\ (x, t, k) &\mapsto C_v(.) \\ (x, t, k) &\mapsto v \end{aligned} \tag{1.1}$$

where $X$ is a set of times to expiry, $T$ is a set of contract tenors and $K$ is a set of strike prices, and where $v$ is the implied volatility. Put differently, $V(x, t, k) = C^{-1}(c; F_{x,t}, k, x)$.

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It is not uncommon, in options modeling, to use appropriate models to relate the price of off-the-money options to at-the-money ones of the same expiry date. Most such models are rich enough to handle smile and skew effects. As a result, the dimensionality of the volatility surface can be reduced by limiting it to at-the-money options. In such case, the implied volatility surface reduces to, $V : X \times T \to \mathbb{C}_v$.

# 2 Forward Stochastic model

The current common options market practice is to model the stochastic behavior of interest rate contracts using the SABR model, namely

$$\begin{aligned} dF &= F^{\beta} \alpha \, dW_F \\ d\alpha &= \alpha \, \nu \, dW_\alpha \\ < dW_F, \, dW_\alpha > &= \rho \end{aligned} \tag{2.1}$$

where $F$ is the risk neutral forward price of the underlying contract, and $dW_F, dW_\alpha$ are the Wiener increments associated with the forward price and its volatility, respectively. Notice that $\alpha$ represents the volatility of the forward contract, while $\nu$ represents the volatility of $\alpha$, and thus termed the vol of vol.

Semi-analytic methods, such as those given by Hagan and by Palout (see Choi and Wu [3] for a full discussion on this topic) are typically used to convert a SABR volatility tuple $(\alpha, \beta, \rho, \nu)$ to an equivalent normal, Black or CEV volatility. If we assume the existence of a function $H : (\alpha, \beta, \rho, \nu; \, x \in X, t \in T, k \in K) \mapsto v$, then simulating the values of $\alpha$ over $X \times T$ should suffice to fully describe the dynamics of the implied volatility surface.

The inverse of the monotonic $H(\alpha|.)$, which we shall denote $H^{-1}(v|.)$, can be used to calculate SABR $\alpha$ from implied volatility, given the reset of the parameters. For a given $x$ and $t$ observable option prices, at a given point in time, for a large enough set of $k$ values can be used to solve for the parameters $\alpha, \beta, \rho$ and $\nu$. The parameters $\beta, \rho$ and $\nu$ tend to remain stable for extended periods of time, and once determined, can be assumed to remain constant during simulation.

For the current work, we will use a CIR variation of the SABR model (2.1). Our model, has the following dynamics

$$\begin{aligned} dF_{x,t} &= F_{x,t}^{\beta_{x,t}} \alpha_{x,t} \, dW_{F_{x,t}} \\ d\alpha_{x,t} &= \theta_{x,t} (\mu_{x,t} - \alpha_{x,t}) \, dt + \nu_{x,t} \sqrt{\alpha}_{x,t} \, dW_{\alpha_{x,t}} \\ < dW_{F_{x,t}}, \, dW_{\alpha_{x,t}} > &= \rho_{x,t} \end{aligned} \tag{2.2}$$

where the index $x, t$ refers to the point in the grid, and $\mu_{x,t}$ and $\theta_{x,t}$ are the process mean and the mean reversion speed, respectively. Unless needed for clarity, we will drop the $x, t$ index in the remainder of the work for simplicity of presentation.

# 3 Parameter Estimation

We rely on historical time series analysis for the estimation of the parameters associated with the model given by (2.2). First, we construct a time series of sufficient length for the implied at-the-money volatility of a given tenor and time to expiry. We then use a semi-analytic model such as the modified Hagan model [5] or the Paulot [7] model to convert the implied volatility time series to a time series for the SABR $\alpha$ of (2.2).

Prykhodko and Ralchenko [8] suggested the following as an unbiased estimator for the diffusion parameter, assuming ergodic behavior and given a time series of forward prices $F_i, i \leq n$, that covers a period of $t_n$ years,

$$\hat{\nu}^2 = \frac{n}{t_n} \frac{\sum_{i<n}(F_i - F_{i-1})^2}{\sum_{i<n} F_i} \,. \tag{3.1}$$

Ben Alaya and Kebaier [2] showed that

$$\frac{1}{t_n} \int_0^{t_n} F_s ds \xrightarrow{\mathbb{P}} \mu \quad \text{as} \quad t \longrightarrow \infty \,. \tag{3.2}$$

They also suggested the following for the consistent estimation of $\theta$

$$\hat{\theta} \;=\; \frac{t_n \left(\log F_n - \log F_0 + \frac{t_n}{2n} S^2 \sum_{i<n} \frac{1}{F_i}\right) - \frac{t_n}{n}(F_n - F_0) \sum_{i<n} \frac{1}{F_i}}{t_n^2 \left( \frac{1}{n^2} \sum_{i<n} \frac{1}{F_i} \sum_{i<n} F_i \;-\; 1 \right)} \,. \tag{3.3}$$

# 4 Option Portfolio

An at-the-money implied volatility surface is usually given with respect to a very large grid made of a list of option expiries along one dimension, and a list of underlying tenors along the other dimension. Reduction of the underlying dimensionality is desired for two main reasons. First, any realistic simulation of an option portfolio has to reflect the natural stochastic dependence inherent when modeling derivatives on highly correlated underlyings. Secondly, full simulation of the entire grid, on a daily basis for long time horizons, results in unjustifiable resource and time consumption.

In the next two subsections, we propose two approaches for complexity reduction. The first approach is based on relating the volatility of all options to the underlying stochastic factors of the forward yield curve itself. The second approach uses Principal Component Analysis (PCA) to achieve dimensionality reduction.

## 4.1 Forward Swap Volatility

Let us start by assuming the following risk neutral dynamics for the instantaneous forward rate $f(t,T)$, which is the rate at time $T$ as observed at time $t$,

$$\begin{aligned} df(t,T) &= \sum_{i=1}^{n} \sigma_i(t,T)\,\bar{\sigma}_i(t,T)\,dt \;+\; \sum_{i=1}^{n} \sigma_i(t,T)\,dW_i(t) \\ \bar{\sigma}_i(t,T) &= \int_t^T \sigma_i(t,u)\,du\,, \end{aligned} \tag{4.1}$$

where $dW_i(t)$ are $n$ standard uncorrelated Wiener increments.

Employing Ito's lemma and the arguments of HJM (see [4]), we get

$$\frac{dP(t,T)}{P(t,T)} = r(t)\,dt \;-\; \sum_{i=1}^{n} \bar{\sigma}_i(t,T)\,dW_i(t)\,, \tag{4.2}$$

where $r(t) = f(t,t)$.

The interest rate forward swap rate $S(t_x,t_s,t_e)$ is the coupon rate of the fixed leg of the swap, which is an $\mathcal{F}_t$-martingale under the annuity measure. The parameters $t_s$ , $t_e$ are the start and end dates of the swap, respectively, and $t_x$ is the rate set date. Let the swap-corresponding annuity be defined as

$$A(t) = \sum_{i=0}^{n} \pi_i\,P(t,T_i) \qquad t \le t_s\,, \tag{4.3}$$

where $n$ is the number of payments, $T_i$ is the $i$-th payment date, $T_0 = t_s, T_n = t_e$ and $\pi_i$ is a calendar factor. The value of $S(t_x,t_s,t_e)$, assuming simple calendar conventions, can be expressed as

$$S(t_x,t_s,t_e) = \frac{P(t_x,t_s) - P(t_x,t_e)}{A(t_x)}\,. \tag{4.4}$$

Let us write the risk neutral dynamics of $S(t_x,t_s,t_e)$ as

$$dS(t,t_s,t_e) = \mu(t)\,dt \;+\; \sigma_{s,e,i}(t)\,dW_i(t)\,. \tag{4.5}$$

Employing Ito's lemma, we have, for $t \leq t_s$

$$
\begin{aligned}
\sigma_{s,e,i}(t) &= \frac{P(t,t_s) - P(t,t_e)}{A^2} \sum_{j=0}^{n} \pi_i P(t,T_j)\, \bar{\sigma}_i(t,T_j) \;-\; \frac{P(t,t_s)}{A(t)}\, \bar{\sigma}_n(t,t_s) + \frac{P(t,t_e)}{A(t)}\, \bar{\sigma}_n(t,t_e) \\
&= \frac{1}{A(t)} \left( \frac{P(t,t_s) - P(t,t_e)}{A(t)} \sum_{j=0}^{n} \pi_i P(t,T_j)\, \bar{\sigma}_i(t,T_j) \;-\; P(t,t_s)\, \bar{\sigma}_n(t,t_s) + \; P(t,t_e)\, \bar{\sigma}_n(t,t_e) \right) \\
&= \frac{1}{A(t)} \left( S(t,t_s,t_e) \sum_{j=0}^{n} \pi_i P(t,T_j)\, \bar{\sigma}_i(t,T_j) \;-\; P(t,t_s)\, \bar{\sigma}_n(t,t_s) + \; P(t,t_e)\, \bar{\sigma}_n(t,t_e) \right)
\end{aligned}
\tag{4.6}
$$

Substituting from Equation (4.1), we get

$$
\begin{aligned}
\sigma_{s,e,i}(t) =& \frac{1}{A(t)} \Bigg( S(t,t_s,t_e) \sum_{j=0}^{n} \pi_i\, P(t,T_j) \int_t^{T_j} \sigma_i(t,u)\, du \\
& - \; P(t,t_s) \int_t^{t_s} \sigma_i(t,u)\, du + \; P(t,t_e) \int_t^{t_e} \sigma_i(t,u)\, du \Bigg) \\
\sigma_{s,e,i}(t) =& \frac{S(t,t_s,t_e)}{A(t)} \sum_{j=0}^{n} \pi_i\, P(t,T_j) \int_t^{T_j} \sigma_i(t,u)\, du \\
& - \; S(t,t_s,t_e) \int_t^{t_s} \sigma_i(t,u)\, du + \frac{P(t,t_e)}{A(t)} \int_{t_s}^{t_e} \sigma_i(t,u)\, du\,.
\end{aligned}
\tag{4.7}
$$

Equation (4.7) represents the forward swap volatility in terms of the terms structure volatilities of the instantaneous forward rates. Using this equation one can construct a set of arbitrage-free swap volatilities, given a set of instant forward volatilities. Similarly, given a set of *consistent* swap volatilities, one should be able to deduce, at least numerically, the instantaneous forward volatilities.

## 4.2 PCA Analysis

Detailed time series analysis of SOFR interest rate swaptions between 2020 and 2026, showed significant correlation. The options included in the dataset included 17 times to expiry and 23 tenors, *i.e.* the $X \times T$ set had a cardinality of 391. It was found out that 76% of the total variance in the normalized (centered and scaled) daily change in the SABR alpha value, was explained by the first principal component. The first three components were sufficient to explain 94% of the total variance. Whereas, 98% was explained by the first five components. These results suggest that the full implied volatility surface can be reliably simulated using a handful of stochastic factors.

Given a historical time series matrix $M$ of $N_h$ rows and $X \times T$ columns, where $N_h$ is the number of observations of $\alpha$ values for each grid point, the SVD (Singular Value Decomposition) of X is given by

$$M = U\ S\ V^\top \tag{4.8}$$

where $M$ is an $N_h \times (X \times T)$ matrix, $U$ an $N_h \times N_h$ matrix, $S$ an $N_h \times (N \times T)$ diagonal matrix of non-negative non-increasing real numbers, and $V$ an $(X \times T) \times (X \times T)$ matrix.

Retaining only the top $p$ principal components, the SABR $\alpha_{x,t}$ of any option $(x, t)$ can be expressed as

$$\begin{aligned} d\alpha_{x,t} =& \theta_{x,t}(\mu_{x,t} - \alpha_{x,t})\, dt \ + \ \nu_{x,t} \sqrt{\alpha}_{x,t} \sum_{i \leq p} \rho_{i,x,t}\, dW_i \\ \rho_{i,x,t} =& \frac{1}{R}\, S_{i,i}\, V_{(x,t),i} \\ R =& \sqrt{\sum_{i \leq p} (S_{i,i} V_{(x,t),i})^2} \end{aligned} \tag{4.9}$$

where $dW_i$ are $p$ Wiener increments for each time step.

With the aid of the semi-analytic formula, $H(\alpha|.)$ of Section 2, each simulated $\alpha_{x,t}$ can be converted to a simulated $V(x, t, k)$, and subsequently, a simulated at-the-money implied volatility surface.

# 5 Monte Carlo Simulation

The nonlinearity involved in the CIR process requires delicate attention to the discretization scheme employed in the Monte Carlo simulation. The algorithms developed by Alfonsi [1] offer a few higher order reliable schemes, which allow a larger simulation time step without loss of accuracy while preserving the non-negativity condition on $\alpha$.

In the current work, we chose the $E(0)$ discretization algorithm specified by Alfonsi [1]. Using our notation, Alfonsi's $E(0)$ is given by

$$\hat{\alpha}_{t_{i+1}} = \hat{\alpha}_{t_i}(1 - \theta\delta) \ + \ \nu\sqrt{\hat{\alpha_{t_i}}}(W_{t_{i+1}} - W_{t_i}) \ + \ \frac{1}{4}\nu^2(W_{t_{i+1}} - W_{t_i})^2 + \delta(\mu\theta - \frac{1}{4}\nu^2) \tag{5.1}$$

where $\delta$ is the simulation time step.

# 6 Arbitrage Testing and Fixing

Johnson and Nonas [6] argued that the only arbitrage constraint that needs to be respected by swaptions on the same rate is the triangular inequality, namely,

$$C(v_1; F_{x,t1}, k, x) + C(v_2; F_{t1+x,t2}, k, t1 + x) \geq C(v_3; F_{x,t1+t2}, k, x)\,. \tag{6.1}$$

It is possible that the simulation generated implied volatility surface violates the condition (6.1). To test for any violation of condition (6.1) and to alleviate any such violation, we offer the following algorithm.

```
    assume a grid of $\alpha(x,t)$ values
    assume a grid of $V(x,t)$ values
    assume a function $H(k)$ that maps $\alpha(x,t)$ to $V(x,t,k)$
    let $h_x[x]$ be a hash table that returns the values $t$ for all valid $\alpha(x,t)$
    let $h_t[t]$ be a hash table that returns the values $x$ for all valid $\alpha(x,t)$
    for $x$ in $X$ (in increasing order):
100     for $t$ in $h_x[x]$ (in increasing order):
            $C1 = C(V(x,t,F_{x,t}))$
            for $t1$ in $h_x[t+x]$ (in increasing order):
                $C2 = C(V(x+t,t1,F_{x,t}))$
                if $\alpha(x,t+t1)$ exists:
                    $C3 = C(V(x,t+t1,F_{x,t}))$
                    if $C3 > C1 + C2$:
                        # arbitrage violation
                        # repair
                        set $\alpha(x,t)$ such that $V(x,t,F_{x,t})$=$V(x,t+t1,F_{x,t})$
                        set $\alpha(x+t,t1)$ such that $V(x+t,t1,F_{x,t})$=$V(x,t+t1,F_{x,t})$
                        goto 100
```

# 7 Backtesting

A deep discussion of several detailed approaches to backtesting are given by Tawfik in [9].

# References


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